

T3 group factors: EFT1 after integrating out F

Active theory: $\text{SM} + S_1 + S_2 + C_{LLS_1S_2}$. The fermion F is absent, so y_1 , y_2 and M_F are not running EFT couplings.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Field representations

run	S_1	S_2	α
T3 – B	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	-1
T3 – C	$(2, -\frac{1}{2})$	$(2, \frac{1}{2})$	-1
T3 – A	$(1, 0)$	$(3, 1)$	0
T3 – D	$(3, -1)$	$(1, 0)$	-2
T3 – E	$(3, 0)$	$(3, 1)$	0

$\beta_{m_{S_1}^2}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\lambda_{S_1} m_{S_1}^2$	6	6	4	8	8
$g_2^2 m_{S_1}^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	0	-12	-12
$g_Y^2 m_{S_1}^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	0	-6	0
$\lambda_{12} m_{S_2}^2$	4	4	6	2	6
$\lambda_{S_1}^{\text{adj}} m_{S_1}^2$	0	0	0	4	4
$\lambda_{12}^{\text{cross}} m_{S_2}^2$	0	0	0	0	4

$\beta_{m_{S_2}^2}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\lambda_{S_2} m_{S_2}^2$	6	6	8	4	8
$g_2^2 m_{S_2}^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	-12	0	-12
$g_Y^2 m_{S_2}^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	-6	0	-6
$\lambda_{12} m_{S_1}^2$	4	4	2	6	6
$\lambda_{S_2}^{\text{adj}} m_{S_2}^2$	0	0	4	0	4
$\lambda_{12}^{\text{cross}} m_{S_1}^2$	0	0	0	0	4

$\beta_{\lambda_{H1}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\lambda_{H1} \text{Tr}(Y_d^\dagger Y_d)$	6	6	6	6	6
$\lambda_{H1} \text{Tr}(Y_e^\dagger Y_e)$	2	2	2	2	2
$\lambda_{H1} \text{Tr}(Y_u^\dagger Y_u)$	6	6	6	6	6
$ \lambda_{T3} ^2$	6	6	12	4	4
$g_2^2 \lambda_{H1}$	-9	-9	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_Y^2 \lambda_{H1}$	-3	-3	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{3}{2}$
$\lambda_H \lambda_{H1}$	6	6	6	6	6
$(\lambda_{H1})^2$	4	4	4	4	4
$\lambda_{H1} \lambda_{S1}$	6	6	4	8	8
$\lambda_{12} \lambda_{H2}$	4	4	6	2	6

g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	0	6	6
g_Y^4	$\frac{3}{4}$	$\frac{3}{4}$	0	3	0
$\left(\lambda_{H1}^{\text{adj}}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	0	2	2
$\lambda_{H1} \lambda_{S1}^{\text{adj}}$	0	0	0	4	4
$\lambda_{H2} \lambda_{12}^{\text{cross}}$	0	0	0	0	4

$\beta_{\lambda_{H2}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\lambda_{H2} \text{Tr}(Y_d^\dagger Y_d)$	6	6	6	6	6
$\lambda_{H2} \text{Tr}(Y_e^\dagger Y_e)$	2	2	2	2	2
$\lambda_{H2} \text{Tr}(Y_u^\dagger Y_u)$	6	6	6	6	6
$ \lambda_{T3} ^2$	6	6	4	12	4
$g_2^2 \lambda_{H2}$	-9	-9	$-\frac{33}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$
$g_Y^2 \lambda_{H2}$	-3	-3	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\lambda_H \lambda_{H2}$	6	6	6	6	6
$(\lambda_{H2})^2$	4	4	4	4	4
$\lambda_{H2} \lambda_{S2}$	6	6	8	4	8
$\lambda_{12} \lambda_{H1}$	4	4	2	6	6
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	6	0	6
g_Y^4	$\frac{3}{4}$	$\frac{3}{4}$	3	0	3
$\left(\lambda_{H2}^{\text{adj}}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	2	0	2
$\lambda_{H2} \lambda_{S2}^{\text{adj}}$	0	0	4	0	4
$\lambda_{H1} \lambda_{12}^{\text{cross}}$	0	0	0	0	4

$\beta_{\lambda_{12}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$ \lambda_{T3} ^2$	3	3	4	4	$\frac{4}{3}$
$g_2^2 \lambda_{12}$	-9	-9	-12	-12	-24
$g_Y^2 \lambda_{12}$	-3	-3	-6	-6	-6
$(\lambda_{12})^2$	4	4	4	4	4
$\lambda_{12} \lambda_{S1}$	6	6	4	8	8
$\lambda_{12} \lambda_{S2}$	6	6	8	4	8
$\lambda_{H1} \lambda_{H2}$	4	4	4	4	4
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	0	0	12
g_Y^4	$\frac{3}{4}$	$\frac{3}{4}$	0	0	0
$\left(\lambda_{12}^{\text{adj}}\right)^2$	$\frac{3}{4}$	$\frac{3}{4}$	0	0	4
$\lambda_{12} \lambda_{S2}^{\text{adj}}$	0	0	4	0	4
$\lambda_{12} \lambda_{S1}^{\text{adj}}$	0	0	0	4	4
$(\lambda_{12}^{\text{cross}})^2$	0	0	0	0	4
$\lambda_{12}^{\text{cross}} \lambda_{S1}^{\text{adj}}$	0	0	0	0	4
$\lambda_{12}^{\text{cross}} \lambda_{S2}^{\text{adj}}$	0	0	0	0	4

$\beta_{\lambda_{T3}}$

$\mathcal{S}$	T3 – B ($\alpha = -1$)	T3 – C ($\alpha = -1$)	T3 – A ($\alpha = 0$)	T3 – D ($\alpha = -2$)	T3 – E ($\alpha = 0$)
$\lambda_{T3} \text{Tr}(Y_d^\dagger Y_d)$	6	6	6	6	6
$\lambda_{T3} \text{Tr}(Y_e^\dagger Y_e)$	2	2	2	2	2
$\lambda_{T3} \text{Tr}(Y_u^\dagger Y_u)$	6	6	6	6	6

$\lambda_{T3} \lambda_H$	2	2	2	2	2
$\lambda_{T3} \lambda_{H1}$	4	4	4	4	4
$\lambda_{T3} \lambda_{H2}$	4	4	4	4	4
$\lambda_{T3} \lambda_{12}$	2	2	2	2	2
$g_2^2 \lambda_{T3}$	-9	-9	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{33}{2}$
$g_Y^2 \lambda_{T3}$	-3	-3	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\lambda_{T3} \lambda_{H1}^{\text{adj}}$	-2	-2	0	-4	-2
$\lambda_{T3} \lambda_{H2}^{\text{adj}}$	2	2	4	0	2
$\lambda_{T3} \lambda_{12}^{\text{adj}}$	$-\frac{1}{2}$	$-\frac{1}{2}$	0	0	2
$\lambda_{T3} \lambda_{12}^{\text{cross}}$	0	0	0	0	-2

Direct Weinberg mixing

The active EFT1 scalar interaction produces

$$16\pi^2 \beta_{C_5} \supset R_W \lambda_{T3} C_{LLS_1S_2}.$$

$\mathcal{G}$	T3 - B ($\alpha = -1$)	T3 - C ($\alpha = -1$)	T3 - A ($\alpha = 0$)	T3 - D ($\alpha = -2$)	T3 - E ($\alpha = 0$)
R_W	$\frac{4\sqrt{3}}{3}$	$-\frac{4}{3}$	$\frac{4\sqrt{3}}{3}$	$\frac{4\sqrt{3}}{3}$	$-\frac{4\sqrt{6}}{3}$

The displayed R_W values use the project's validated Wigner-6j reduction and inherited tensor convention. The overall tensor-level normalization/phase has not yet been independently derived from first principles.

F-threshold matching tensor samples

The component bases have representation-dependent dimensions, so the raw tensor entries are not placed in the cross-run beta-function tables.

T3 - B ($\alpha = -1$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1155}	$\frac{\overline{y_1}^2}{4MF}$
C_{1156}	$-\frac{i\overline{y_1}^2}{4MF}$
C_{11511}	$\frac{\overline{y_1}y_2}{4MF}$
C_{11512}	$\frac{i\overline{y_1}y_2}{4MF}$
C_{1165}	$-\frac{i\overline{y_1}^2}{4MF}$
C_{1166}	$-\frac{\overline{y_1}^2}{4MF}$
C_{11611}	$-\frac{i\overline{y_1}y_2}{4MF}$
C_{11612}	$\frac{\overline{y_1}y_2}{4MF}$
C_{11115}	$\frac{\overline{y_1}y_2}{4MF}$
C_{11116}	$-\frac{i\overline{y_1}y_2}{4MF}$

T3 - C ($\alpha = -1$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1155}	$\frac{\sqrt{3}\overline{y_1}^2}{12MF}$
C_{1156}	$-\frac{\sqrt{3}i\overline{y_1}^2}{12MF}$
C_{11511}	$-\frac{\sqrt{3}\overline{y_1}y_2}{12MF}$
C_{11512}	$-\frac{\sqrt{3}i\overline{y_1}y_2}{12MF}$

C_{1165}	$-\frac{\sqrt{3}i\overline{y_1}^2}{12MF}$
C_{1166}	$-\frac{\sqrt{3}\overline{y_1}^2}{12MF}$
C_{11611}	$\frac{\sqrt{3}i\overline{y_1}y_2}{12MF}$
C_{11612}	$-\frac{\sqrt{3}\overline{y_1}y_2}{12MF}$
C_{11115}	$-\frac{\sqrt{3}\overline{y_1}y_2}{12MF}$
C_{11116}	$\frac{\sqrt{3}i\overline{y_1}y_2}{12MF}$

T3 – A ($\alpha = 0$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{1157}	$\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{1158}	$\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$
C_{11511}	$\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$
C_{11512}	$-\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{1167}	$-\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$
C_{1168}	$\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{11611}	$\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{11612}	$\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$
C_{1175}	$\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{1176}	$-\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$

T3 – D ($\alpha = -2$)

48 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11511}	$\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{11512}	$\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$
C_{11611}	$-\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$
C_{11612}	$\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{11911}	$\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$
C_{11912}	$-\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{111011}	$\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{111012}	$\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$
C_{11115}	$\frac{3^{\frac{3}{4}}\overline{y_1}y_2}{12MF}$
C_{11116}	$-\frac{3^{\frac{3}{4}}i\overline{y_1}y_2}{12MF}$

T3 – E ($\alpha = 0$)

96 non-zero real-basis tensor components reconstructed; showing the first 10.

C_{ijab}	matched coefficient
C_{11513}	$\frac{\sqrt{3}\overline{y_1}y_2}{12MF}$
C_{11514}	$\frac{\sqrt{3}i\overline{y_1}y_2}{12MF}$
C_{11613}	$-\frac{\sqrt{3}i\overline{y_1}y_2}{12MF}$
C_{11614}	$\frac{\sqrt{3}\overline{y_1}y_2}{12MF}$

$$\begin{array}{lcl}
C_{11711} & - & \frac{\sqrt{3}y_1y_2}{12MF} \\
C_{11712} & - & \frac{\sqrt{3}iy_1y_2}{12MF} \\
C_{11715} & - & \frac{\sqrt{3}iy_1y_2}{12MF} \\
C_{11716} & & \frac{\sqrt{3}y_1y_2}{12MF} \\
C_{11811} & & \frac{\sqrt{3}iy_1y_2}{12MF} \\
C_{11812} & - & \frac{\sqrt{3}y_1y_2}{12MF}
\end{array}$$

Next threshold

At the next threshold, S_1 and S_2 are integrated out. Their representation dependence is then absorbed into the matching value of C_5 ; below that threshold the SMEFT Weinberg-operator anomalous dimension is universal.